

BITKAUN V8 FORENSICS // COMMAND & ID CHEAT SHEET

Smart India Hackathon 2026 • Air-Gapped Dual-Stream Bitcoin AML Forensics Engine

Verified Working Invocations across OS CLI (Ubuntu WSL / Windows) and Web Terminal

Target Backend: http://localhost:8000	Indexed Tx Count: 294,717 Transactions	AI Models: XGBoost + Isolation Forest
Web Terminal: http://localhost:5173	Unique Wallets: 984,132 Addresses	Case Storage: %LOCALAPPDATA%\BitKaun

Command & Syntax	Working ID / Invocations (2 Verified Examples)	Forensic Purpose & Demonstrated Output
search search <query>	1. search AS49870 2. search 152.60.31.67	1. Unmasks all mixing relay nodes operating under ASN 49870. 2. Unmasks origin node (Residential, SC, 482 ms propagation latency).
correlate correlate upload	1. correlate 2. ingest sample-pair	1. Opens Dual-Stream Ledger & P2P Telemetry correlation view in Web UI. 2. Correlates 100 on-chain UTXO rows with P2P frames; runs instant SHAP scoring.
inspect inspect <txid addr>	1. inspect 932451115 2. inspect 881920041	1. Massive 10,034 BTC peeling tx, 1-in 2-out, Green Floid LLC ISP. 2. Ransomware extortion payment with multi-party victim consolidation.
dossier dossier <txid> [--save]	1. dossier 932451115 --save 2. dossier 881920041 --save	1. Generates FIU-IND prosecution summary; saves JSON to active case. 2. Section 65B court evidence package for ransomware syndicate.
flow flow <txid>	1. flow 932451115 2. flow 324013641	1. Decomposes peeling split: 10,034 BTC into change address and peel hop. 2. Decomposes CoinJoin mixer pool: multi-input equal-split distribution.
anomaly anomaly <scenario_id>	1. anomaly mixing_05121 2. anomaly peeling_chain_04606	1. Isolation Forest score: 58.18 / 100 [MEDIUM] (Raw: -0.5154). 2. Isolation Forest score: 74.24 / 100 [HIGH] unusualness deviation.
graph graph <scenario_id>	1. graph peeling_chain_04606 2. graph mixing_05121	1. Web UI: 3D WebGL camera focuses on linear peeling chain trail. 2. CLI: Prints node/edge counts & hub wallets for CoinJoin mixing pool.
communities communities <scenario_id>	1. communities peeling_chain_04601 2. communities mixing_05121	1. Greedy Modularity detects coordinator vs change wallet syndicates. 2. Separates pre-mix depositor clusters from post-mix clean recipients.

taint taint <seed_address>	1. taint 14c55RuJwJivdZmEjpTDJNjcAfvLdkaH 2. taint 1c9Tm1sAir7vV5mn7e3zZwo6eDZZd16RHZ	1. Computes forward dirty coin decay from peeling syndicate master head. 2. Mathematical FIFO taint propagation across downstream change wallets.
trace trace <src> <dst>	1. trace 1c9Tm1sAir7vV5mn7e3zZwo6eDZZd16RHZ 1Cc18UpLkcS5FcZsKuPxKWpv1fT 2. trace 1Vo1CYJGZqnHYk2TKjNiuawkTZgZ47n 1cUUSwM6WHg23QmLUXPGdWWY2XHZL68W4	1. Multi-hop BFS velocity trace across peeling chain 04601. 2. Direct fund routing path across peeling chain 04602 to cashout exit.
tor tor [txid]	1. tor 932451115 2. tor 324013641	1. Shannon timing entropy test: Tor Negative, 482 ms Clearnet propagation. 2. Evaluates gossip broadcast delay and timing jitter for mixer node.
ingest ingest sample <type>	1. ingest sample peeling 2. ingest sample ransomware	1. Injects synthetic peeling chain with instant real-time XGBoost ML scoring. 2. Injects high-threat extortion transaction into live in-memory graph.
scenarios scenarios [prefix] [page]	1. scenarios peel 1 2. scenarios mix 1	1. Paginated directory filtering all active peeling chain clusters. 2. Directory of all CoinJoin / Wasabi mixing scenario clusters.
cases / cd cases cd <case> cd ..	1. cases 2. cd test3 (Exit: cd ..)	1. Lists all cases stored in private local AppData with artifact counts. 2. Activates case 'test3' for evidence isolation. 'cd ..' returns to global.
benchmark / telemetry benchmark telemetry	1. benchmark 2. telemetry	1. V8 Frozen Scorecard: 93.9% F1, 0.42 ms sub-millisecond AI latency. 2. Global P2P infrastructure census: Tor vs Residential vs Datacenter nodes.

HACKATHON PRESENTATION & EVALUATION CHEAT SHEET (SIH 2026)

- 1. Latency Breakdown:** AI ML Inference is **0.42 ms** (420 microseconds) per transaction. P2P propagation latency is **<500 ms** on clearnet vs **800–8,500 ms** on Tor relays.
- 2. Why Models Don't Self-Learn Online:** Prevents adversarial poisoning by criminals and guarantees court admissibility under Section 65B of the Indian Evidence Act with a certified frozen benchmark version.
- 3. Dual-Stream Fusion:** BitKaun does not just look at on-chain UTXO graphs; it fuses pre-block Layer-0 gossip network telemetry (Relay IP, ASN, Tor exit status) with on-chain ledgers.
- 4. Flexible Schema Ingestion:** Supports user CSV uploads with arbitrary headers (e.g. *From_Address, TxHash, Value*) auto-mapped with zero manual formatting.